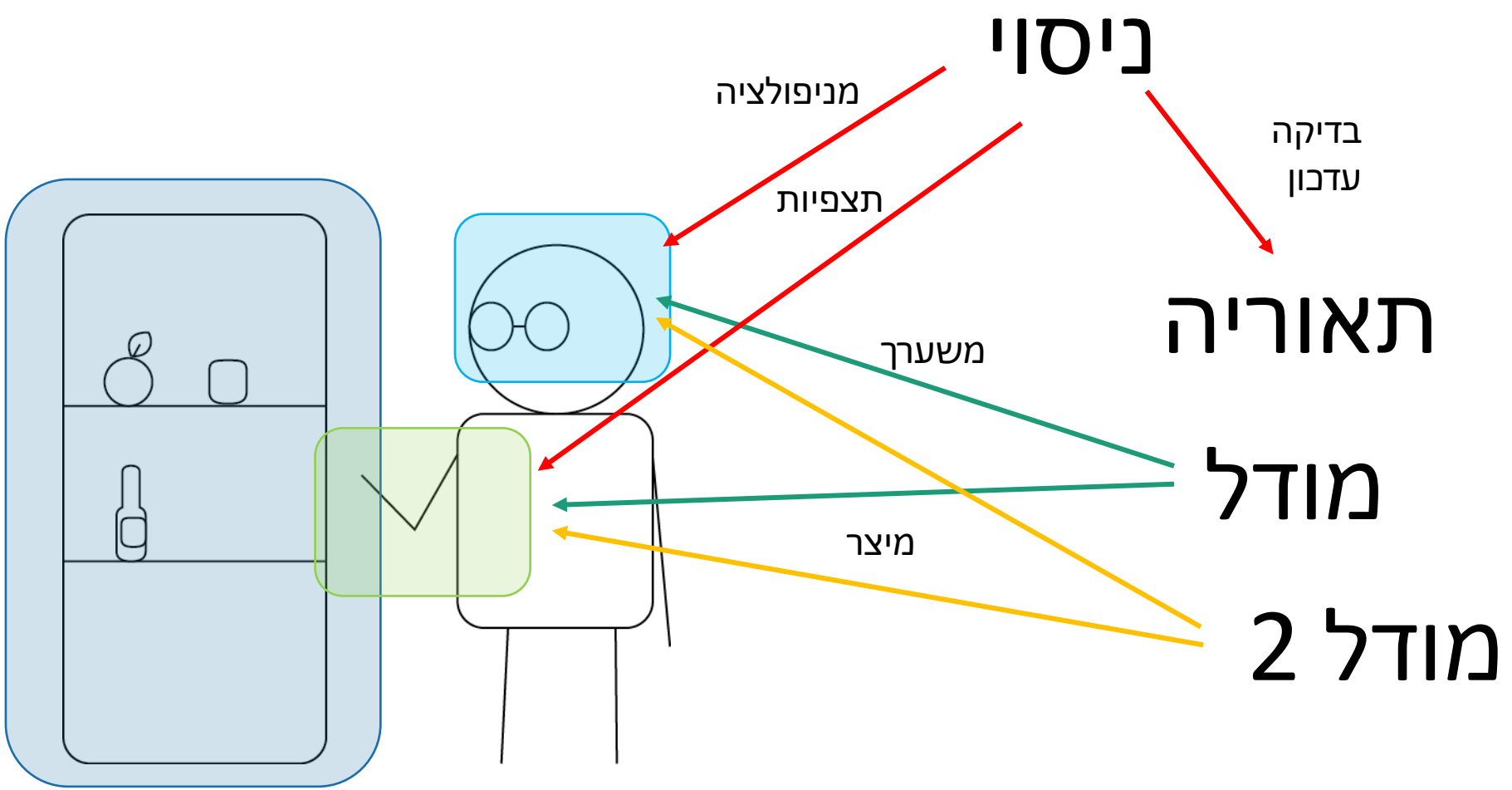
השוואת מודלים

פרק 10 בספר

Model Comparison

- What is the best model?
- How do we compare and evaluate model fitting?
- What are the problems one may encounter?

מידול של תהליכים קוגניטיביים



איך מחליטים איזה מודל יותר טוב?

גישת הנתונים – האם מודל אחד מתאים טוב יותר לנתונים מאשר מודל אחר?

בגישה זו מתמקדים בנתונים שעומדים לפנינו, תוצאה של הנתונים שאספנו מהנבדקים הספציפיים שלנו, בניסוי הספציפי שבנינו.

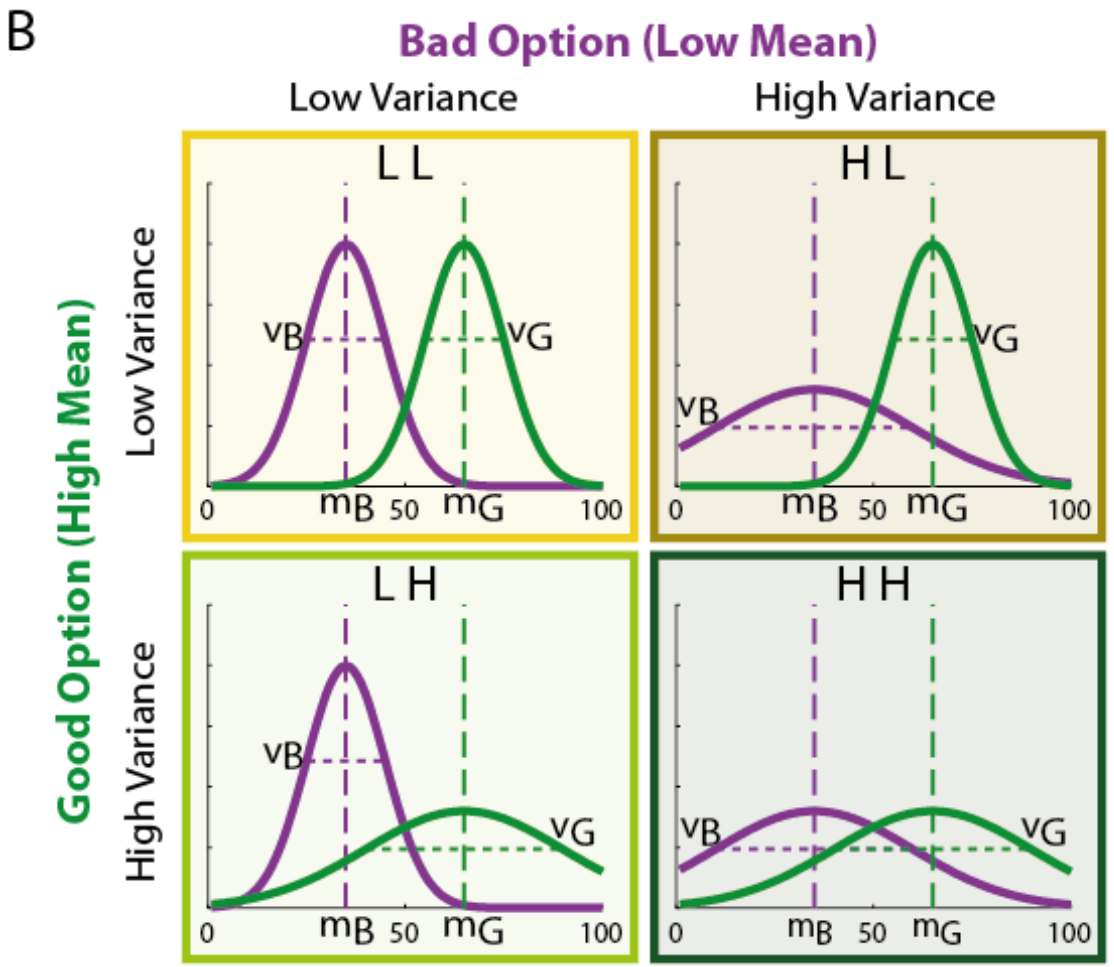
השוואת מודלים

- יכולים להיות יותר ממודל אחד של התהליך הקוגניטיבי שמוביל להתנהגות.

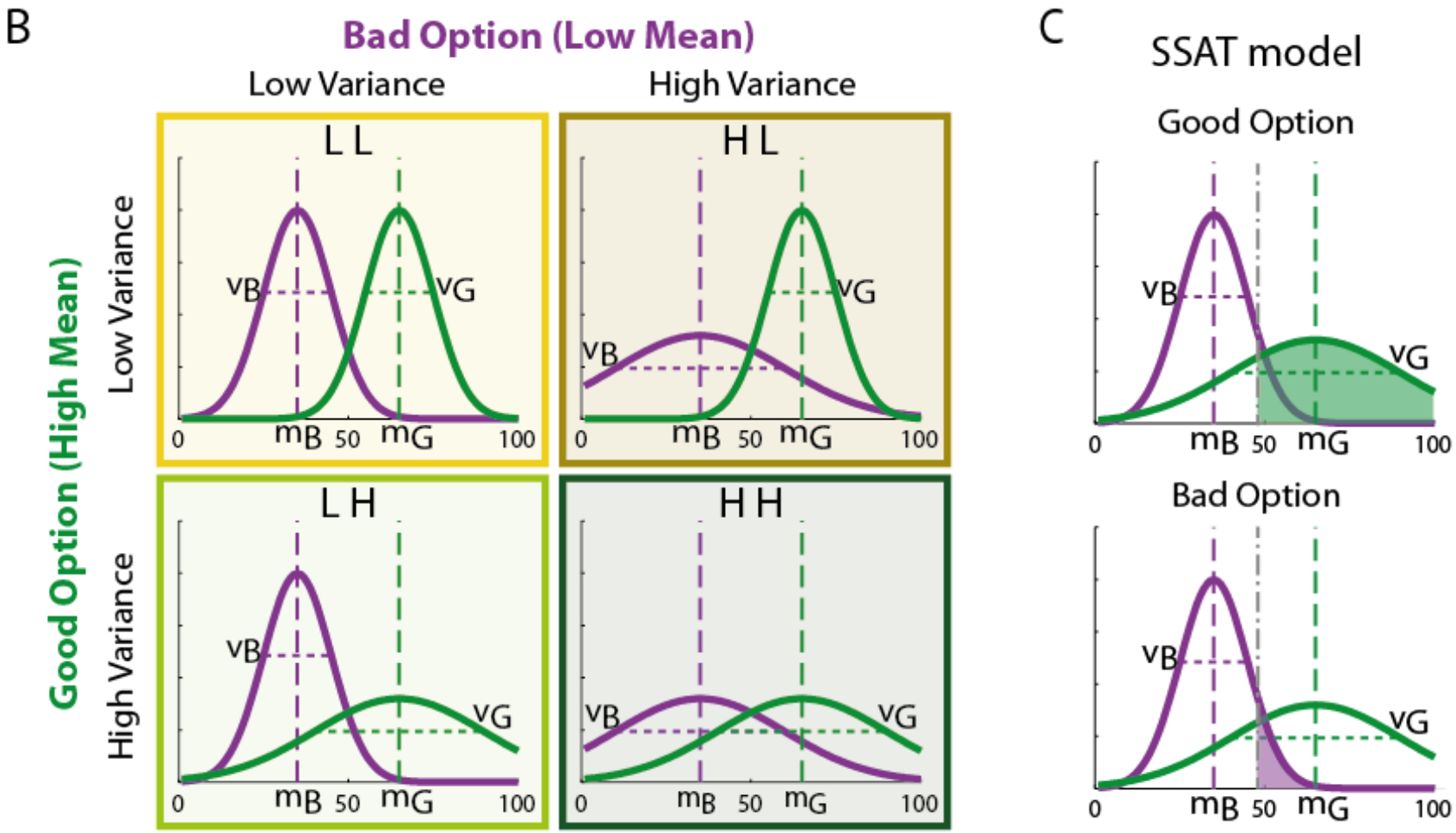


http://urihertz.net/BanditConfDemo/

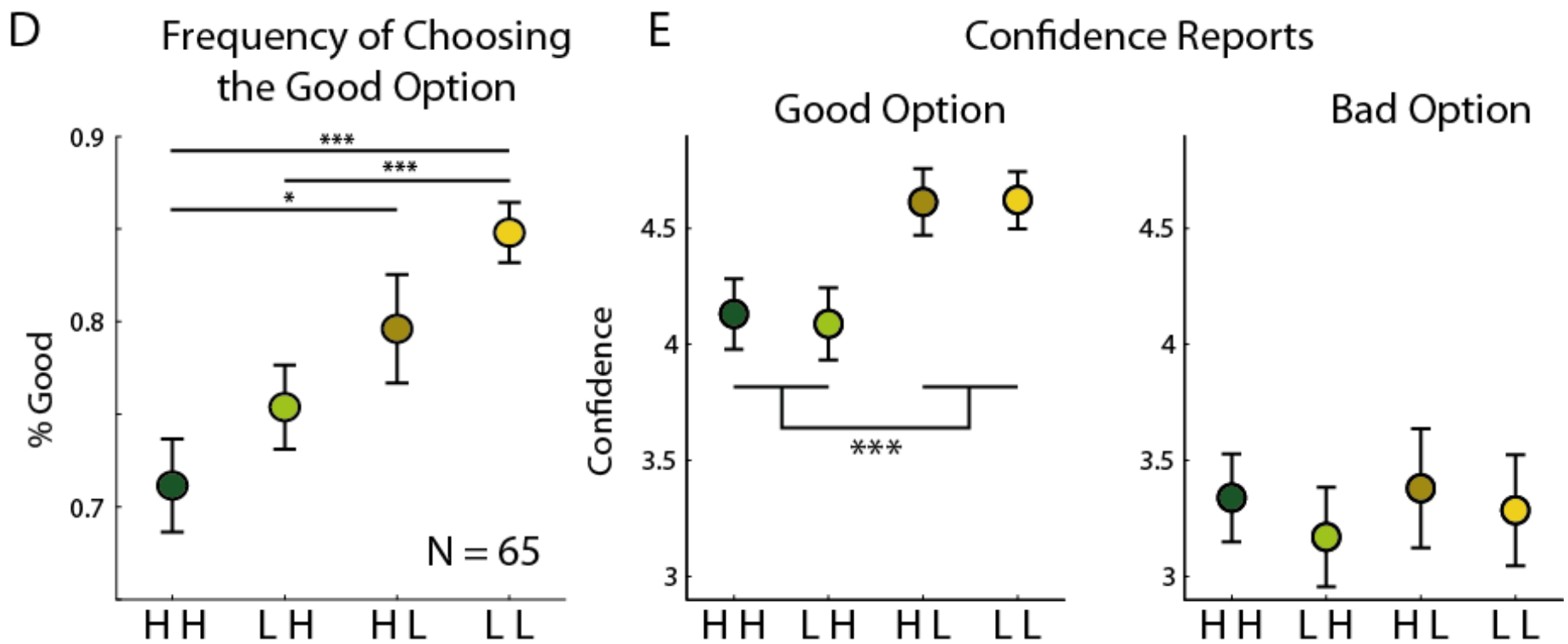
השוואת מודלים



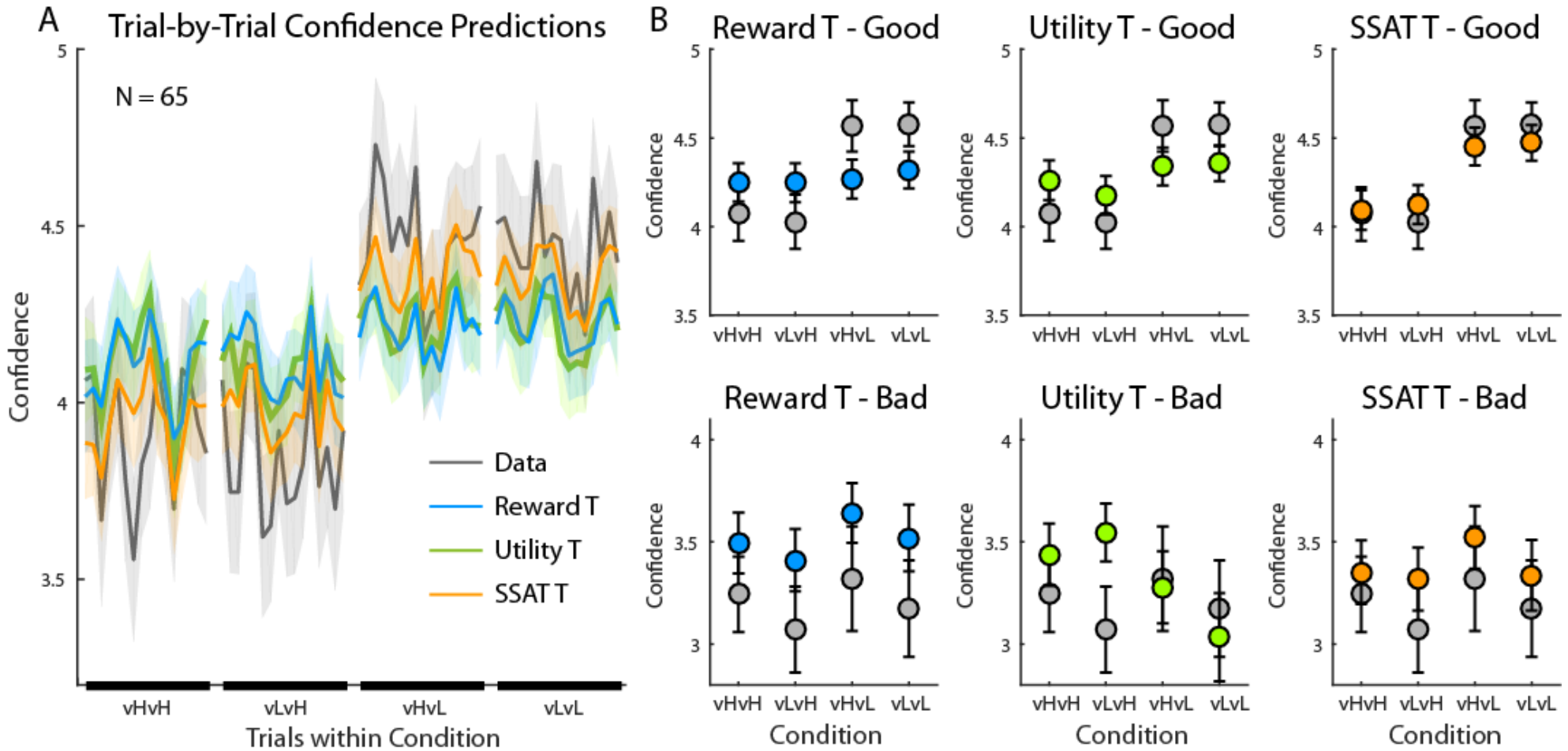
השוואת מודלים



השוואת מודלים

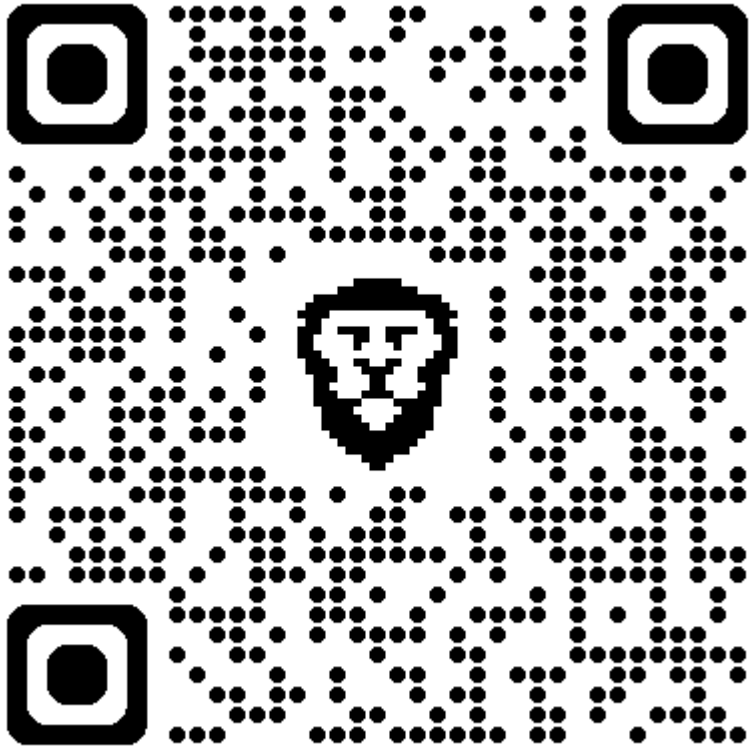


השוואת מודלים



Let’s have another look at our model fitter

- https://sdl-exp.site/CompCogCourse/Lecture7Optim/QuadraticFit.html
- Try to see which model gives you the best fit – linear or quadratic.
- What seems to be the difference in fitting error?



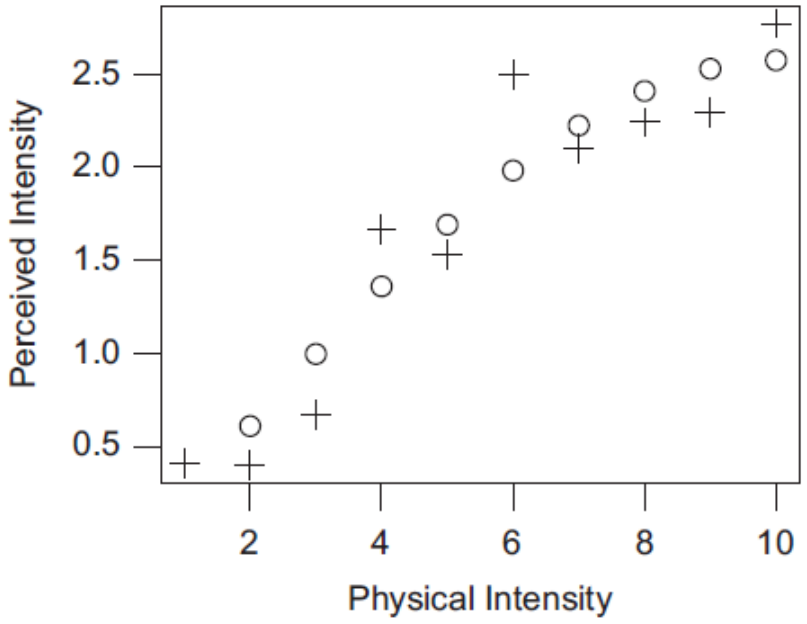
The model with the lowest error is the best!

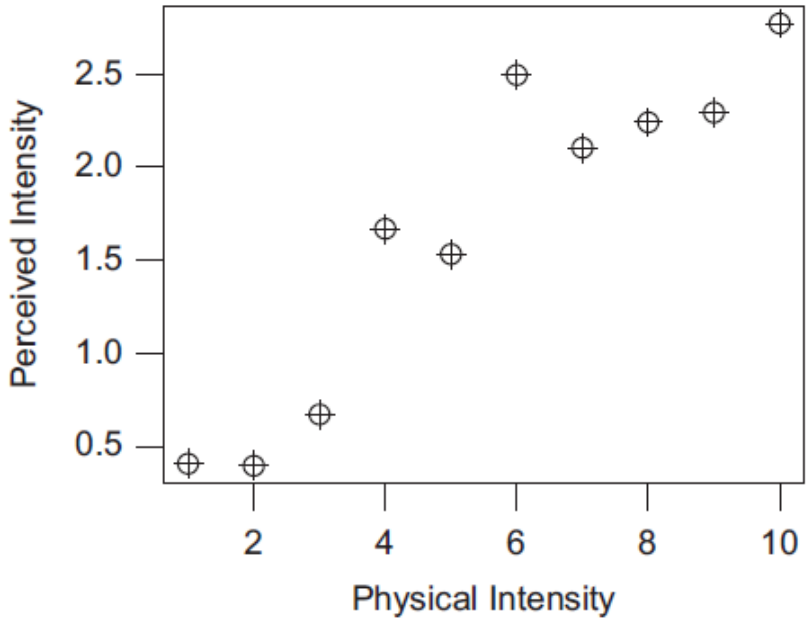
- Or is it?
- Remember – our data is noisy, and this means that sometimes the real model is not the one giving the best fit to the data we just observed.
- https://sdl-exp.site/CompCogCourse/Lecture75OverFit/Noise.html



איך מחליטים איזה מודל יותר טוב?

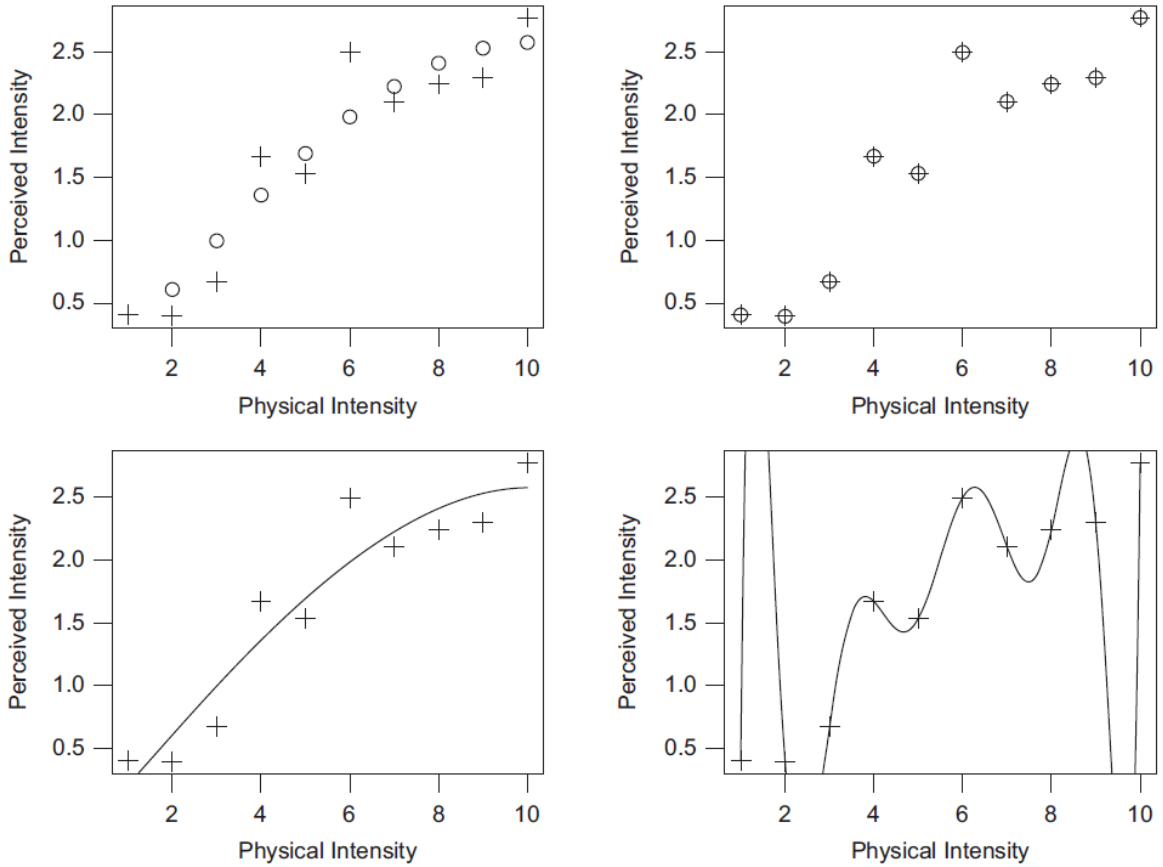
- יש בעיה עם הגישה הזו.
- דוגמה – התאמה של מודל פולינומיאלי להתנהגות.





$$m = \beta_0 + \beta_1 s + \beta_2 s^2 + \beta_3 s^3 + \cdots + \beta_k s^k$$

מודל פולינמיאלי



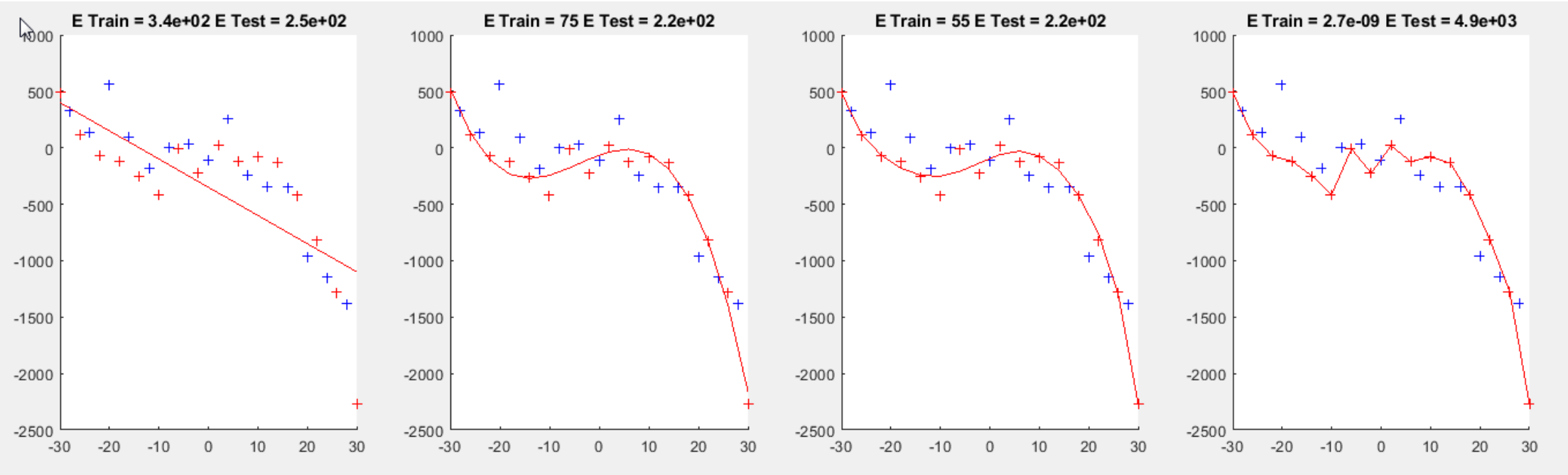
מה הפרדיקציות שהמודלים השונים נותנים לערכי X שונים?

האם יש כאן בעיה?

Overfitting

- A model with more parameters is more flexible, and can capture our data with smaller error.
- But…
- Noise? Generality? Hidden assumptions?

Over-Fitting

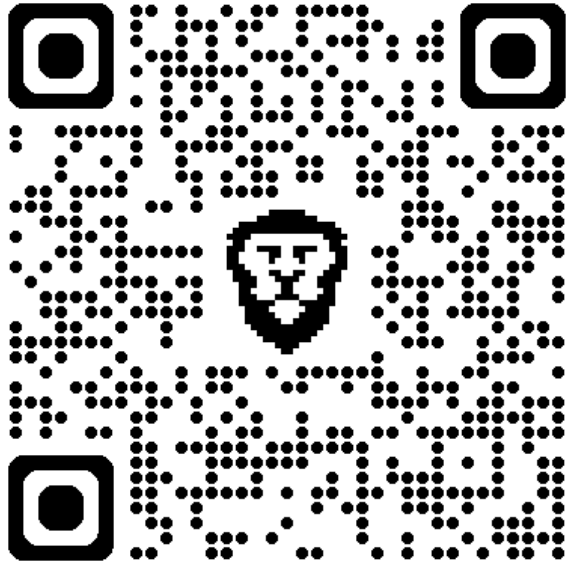


Capturing overfitting – models' bias and variance

- Two measures of models are their bias and variance.
- These refer to values that one can calculated when fitting the same model many times to simulated data.
- The bias is the error – the distance of model estimations from the data points in each run.
- Variance – the differences in model parameters when fitted to different runs of the same model.

Let’s try it…

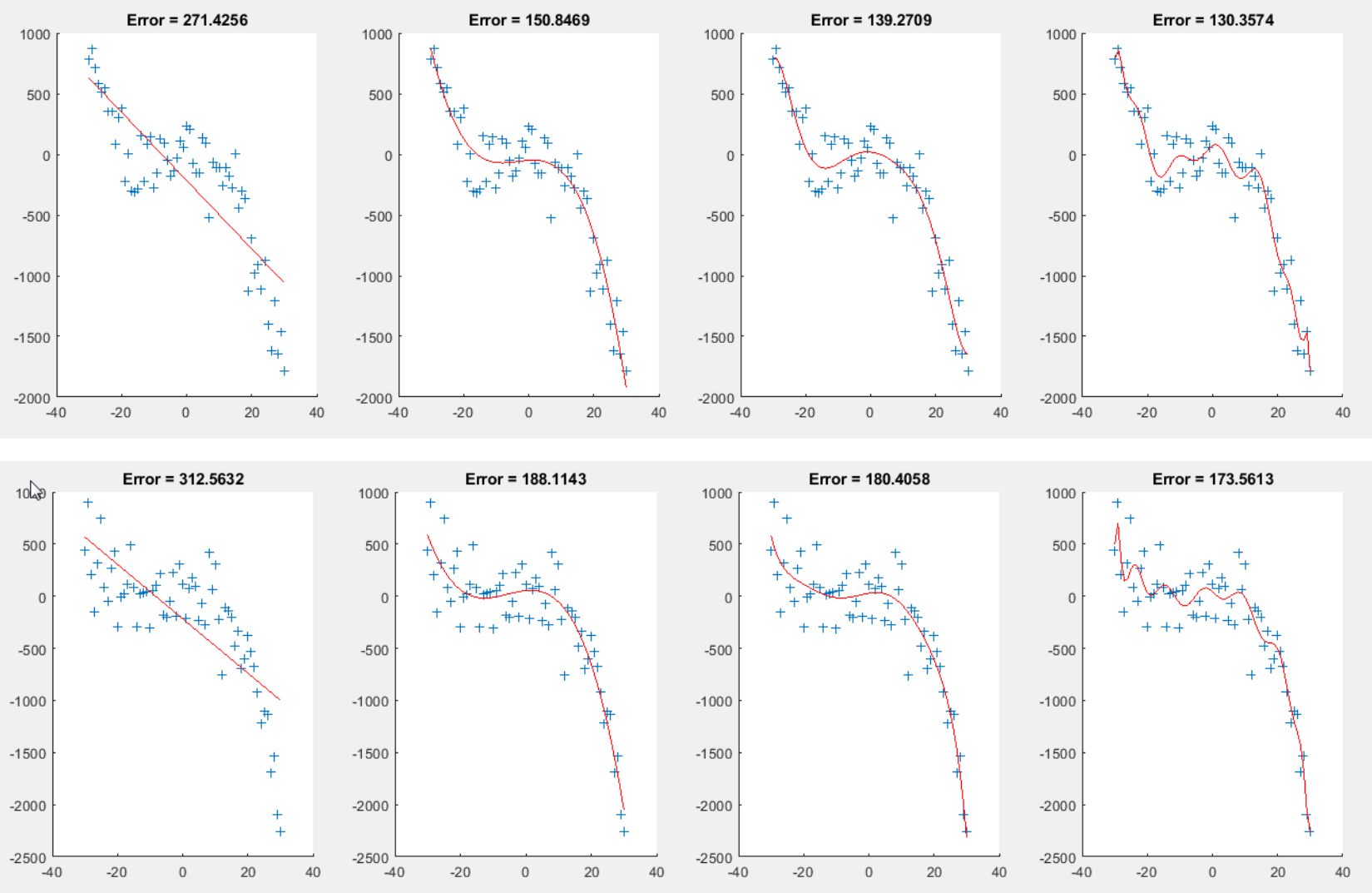
- Generate five model fits for each type of model.
- Save the data and the final report.
- https://sdl-exp.site/CompCogCourse/Lecture75OverFit/OverFit.html



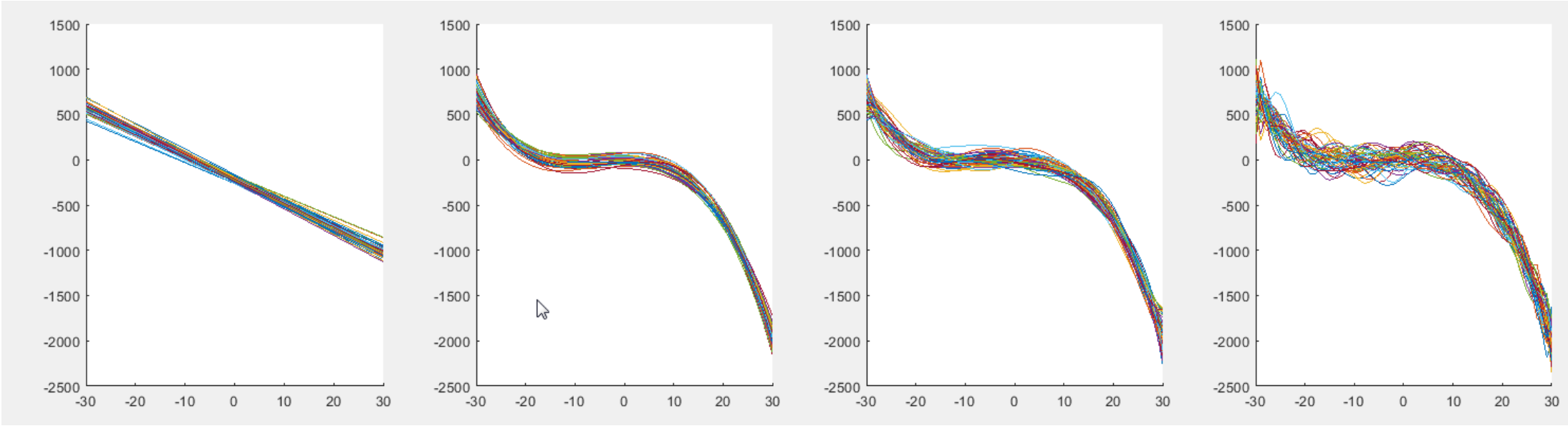
How stable were your estimates?

- Did you need to make a lot of adjustments between fits?
- How much did the noise affect the fitting procedure?
- Which model had the highest bias?
- Which model had the highest variance?

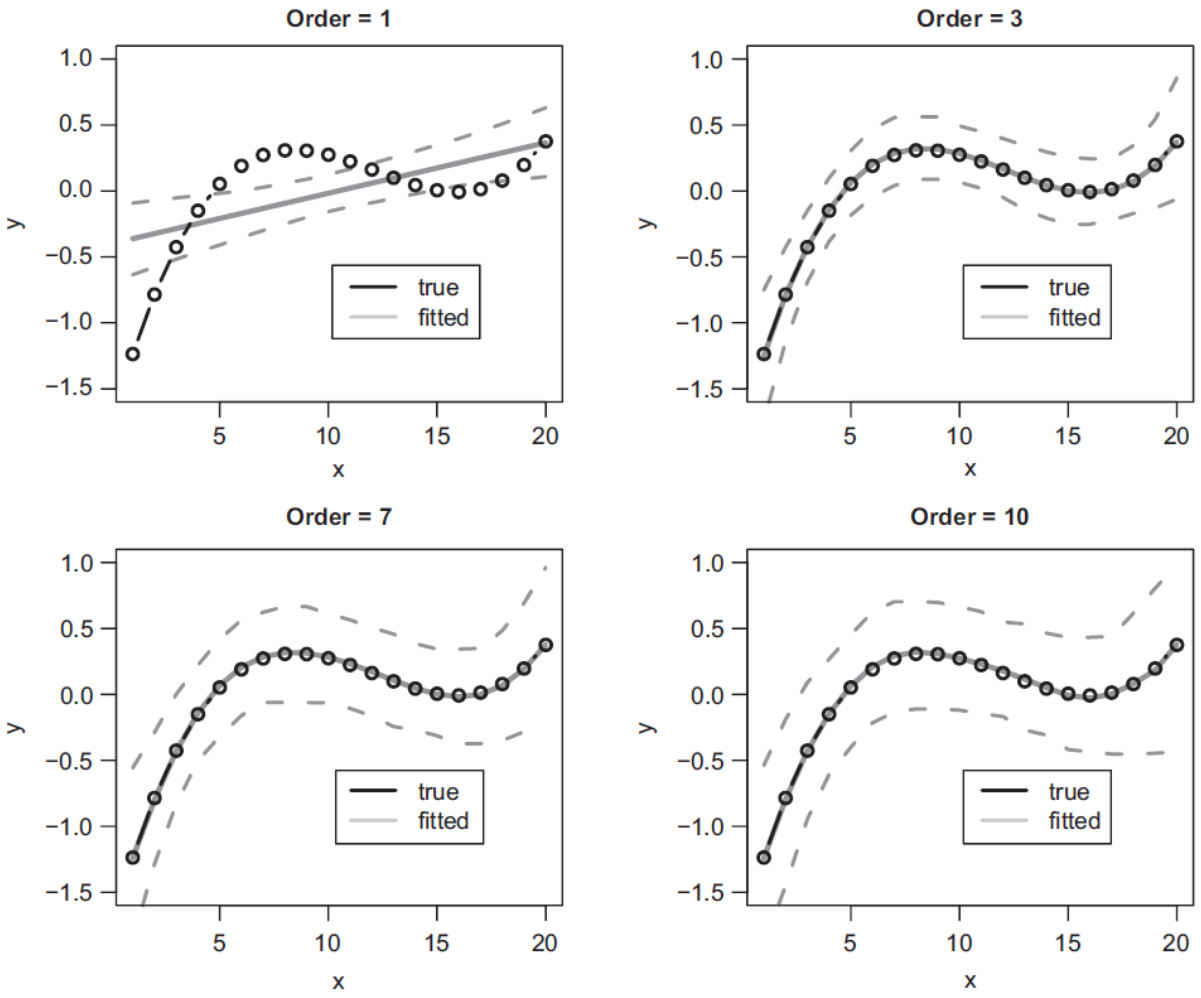
Bias and Variance



Bias and Variance



Bias vs variance



$$y_i = f(x_i) + \epsilon_i$$

$$f(x) = x - x^2 + x^3$$

$$\epsilon \sim N(0, 0.5)$$

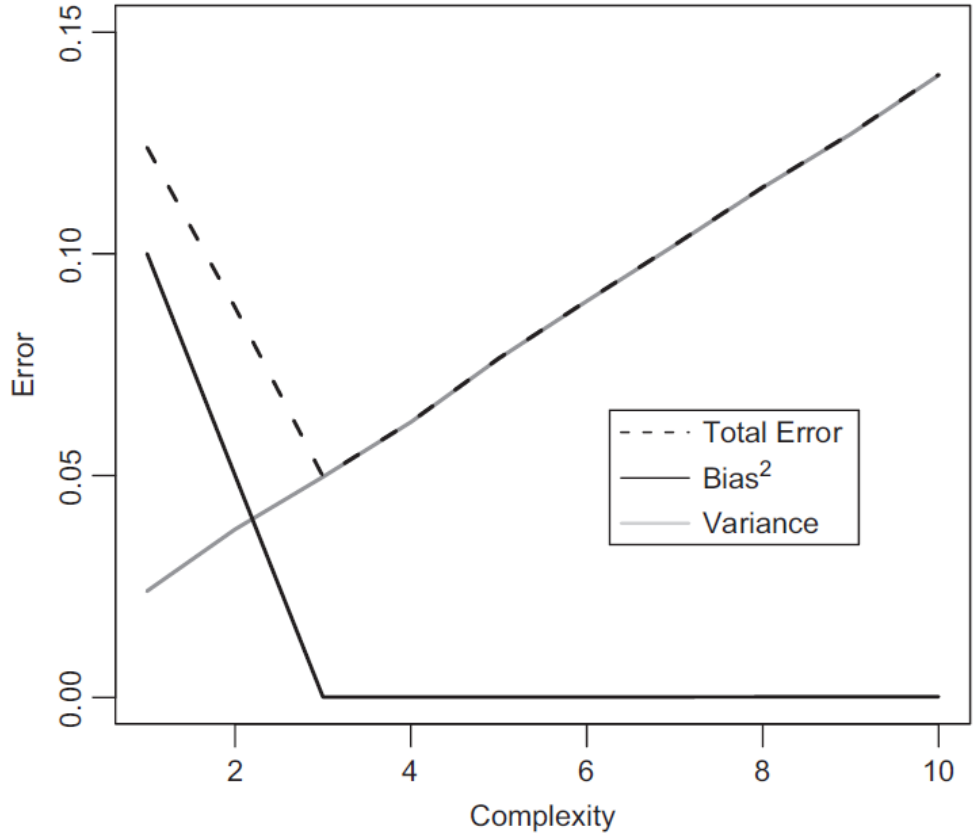
Bias = השגיאה הריבועית

Variance = השונות בהצגות שונות של המודל

ככל שהמודל מורכב יותר השונות שלו בהרצות שונות גדלה.

ככל שהמודל מורכב יותר ההטייה שלו קטנה.

Bias vs variance



$$y_i = f(x_i) + \epsilon_i$$

$$f(x) = x - x^2 + x^3$$

$$\epsilon \sim N(0,0.5)$$

Bias = השגיאה הריבועית
Variance = השונות בהרצות שונות של המודל

ככל שהמודל מורכב יותר השונות שלו בהרצות שונות גדלה.
ככל שהמודל מורכב יותר ההטייה שלו קטנה.

We want a model that has a good balance between bias and variance

- Because we don’t want a model that fits perfectly the noise in our data.
- We want a model that estimates the signal – the model underlying our noisy data.

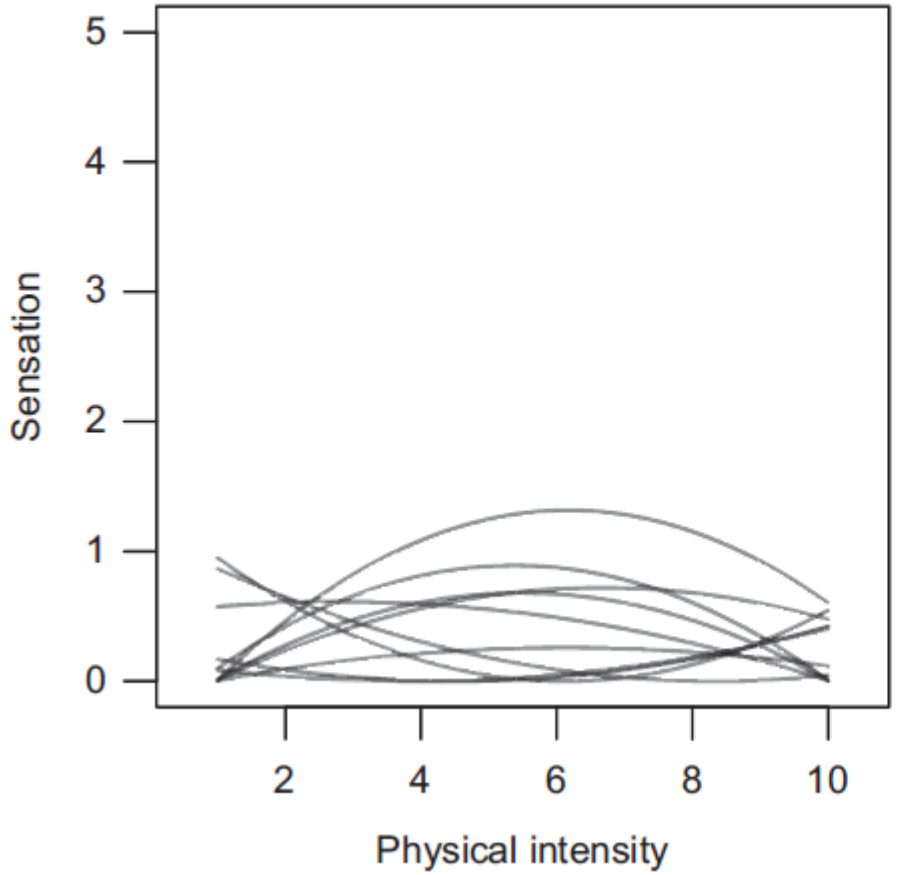
More Data!!

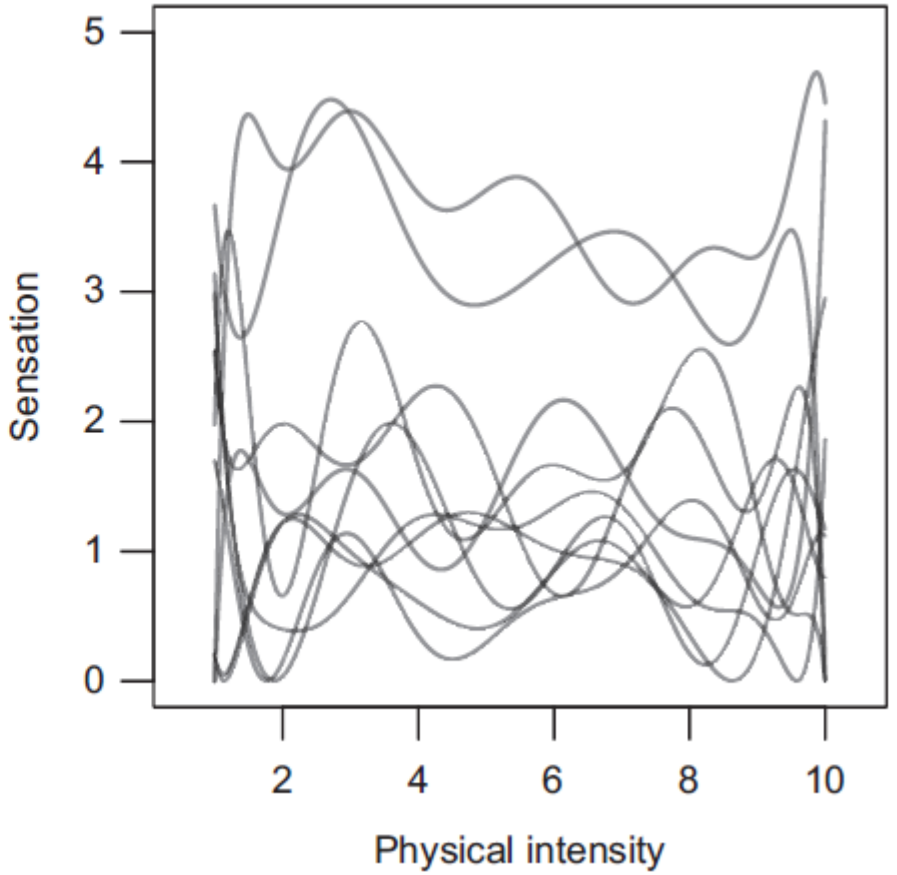
- One way to deal with it is to have a lot of data!
- This is sometimes called the **curse of dimensionality**.
- Let's try fitting our model with only 3 data points, and with 50 data points. When is it more stable? Which fit is more reliable? Which one is more likely to be the 'real' function?

Less noise!!

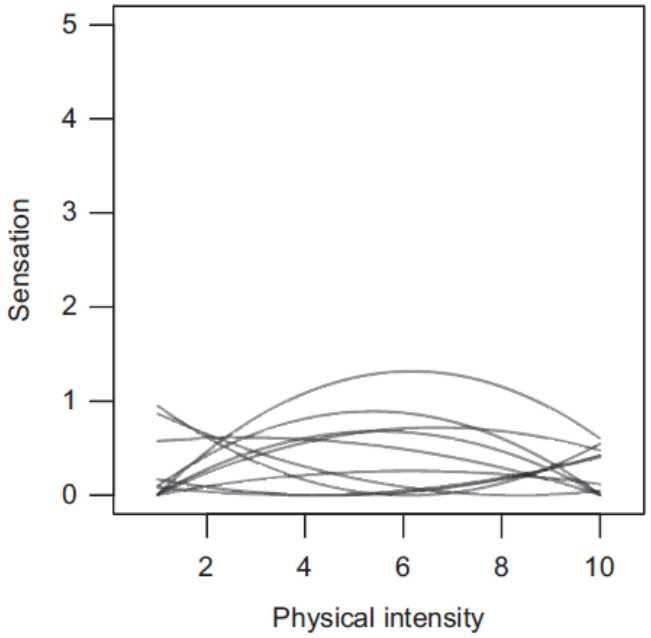
- We can also try to reduce the noise – in this way we the best fit is not capturing noise, but real variations in the data.
- How can we reduce noise?
- **Noise ceiling** - the maximum accuracy a model can have (or the lowest error term). A model shouldn't be more accurate than the noise in the data, otherwise we are over fitting.
- Try playing with the noise term – how does it affect fitting error? Consistency? Curse of dimensionality?

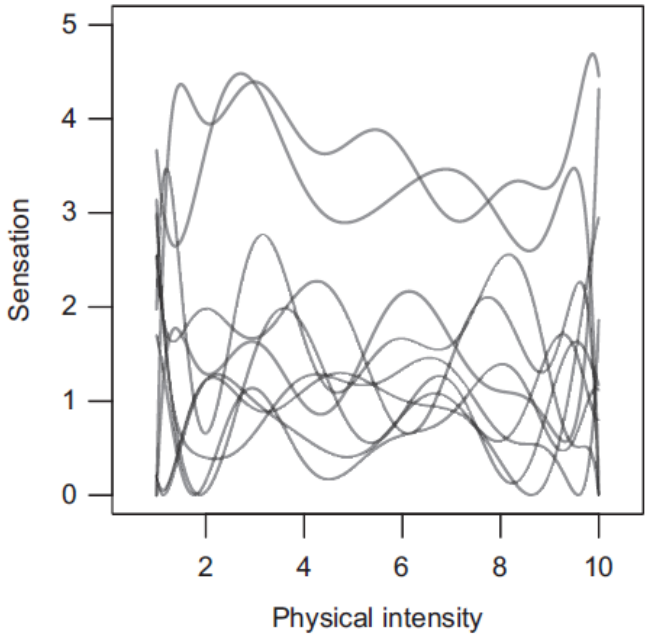
Model Complexity/Flexibility





Model Complexity/Flexibility





- מודלים מורכבים מכילים יותר פרמטרים.
- מסוגלים לבסות מגוון גדול של התנהגויות.
- מסוגלים להכיל מגוון של ערכים בפרמטרים שלהם.

- מודלים כאלה ניתן להתאים למגוון של נבדקים וניסויים.

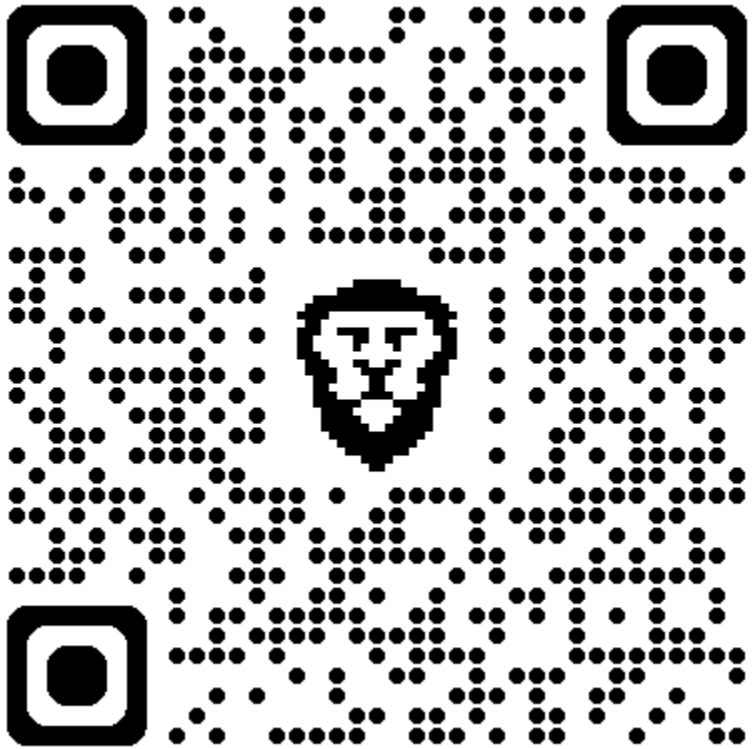
Cross-Validation

- A common way to avoid over-fitting to the noise, is cross-validation.
- We take our data and split it to training set and test set.
- We fit the model to the training set, and evaluate the model (error) on the test set.
- When comparing multiple models, we can compare their test-set error instead of training-set errors.

- In many cases we use this dual split. However, as data is expensive, we can switch the training and test sets, and have two estimations of our model and average them.

Let’s try…

- Try the 2-way cross validation, with different models.
- https://sdl-exp.site/CompCogCourse/Lecture75OverFit/CrossValidation.html

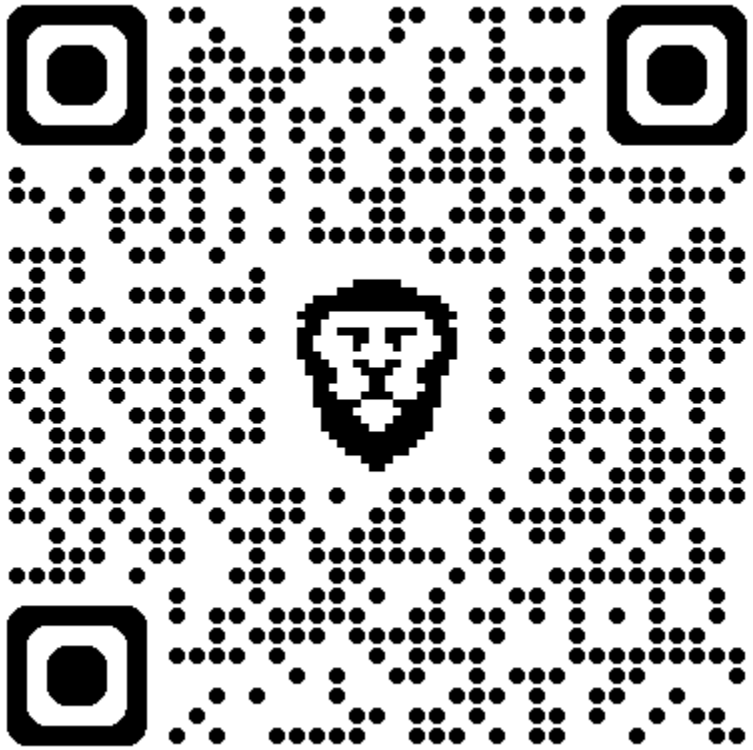


Multiple folds

- Here again, more data gives better perdictions.
- If we have a limited dataset, we can be more economical in our cross-validation – use smaller test windows.
- K-fold cross validation – split the data to k sets, use the one as test and the rest as training set. Do it K times, and average the results (you can also test the variance).
- The most extreme is **leave one out (loo)**.

Let’s try…

- Compare 2-fold to 3/4-fold and 6 fold cross validations, with different models.
- https://sdl-exp.site/CompCogCourse/Lecture75OverFit/CrossValidation.html



Cross-validation can guard us from overfitting!

- Still – depend on noise and number of datapoints.

We want to balance error and generalizability

- How good it fits the current data.
- How good it fits new unseen datasets.

- Adding free parameters can help fitting our current data
- But be detrimental to generalization if there is noise

- Occam's razor heuristic – always prefer a model with less parameters.
- Even penalize models for every additional parameter!